Heaven's Light Is Our Guide

A COMPARATIVE STUDY OF OPINION MINING BASED ON DEEP AND TRADITIONAL MACHINE LEARNING APPROACHES

A Thesis Submitted in Partial

fulfillment for the requirement of the

degree of
Bachelor of Science
in
Electrical & Computer Engineering
by

Mahammed Kamruzzaman

Roll: 1510036

to the

Department of Electrical & Computer Engineering

Rajshahi University of Engineering & Technology

February, 2021

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Acknowledgement

This thesis has been submitted to the Department of Electrical and Computer Engineering of Rajshahi University of Engineering & Technology (RUET), Rajshahi-6204, Bangladesh, for the partial fulfillment of the requirements for the degree of B.Sc. Engineering in Electrical & Computer Engineering. Thesis title regards to " **A COMPARATIVE STUDY OF OPINION MINING BASED ON DEEP AND TRADITIONAL MACHINE LEARNING APPROACHES**".

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CERTIFICATE

This is to certify that the thesis entitled " ***A COMPARATIVE STUDY OF OPINION MINING BASED ON DEEP AND TRADITIONAL MACHINE LEARNING APPROACHES*** " by Mahammed Kamruzzaman (Roll No. 1510036), has been carried out under my direct supervision. To the best of my knowledge, this thesis is an original one and has not been submitted anywhere for any degree or diploma.

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CERTIFICATE

This is to certify that the thesis entitled " ***A COMPARATIVE STUDY OF OPINION MINING BASED ON DEEP AND TRADITIONAL MACHINE LEARNING APPROACHES*** " has been corrected according to my suggestion and guidance as an external. The quality of the thesis is satisfactory.

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Declaration

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List of Symbols

Symbol	Meaning
n(t)	No. of times the term t arises in the document
N(t)	Total no. of terms
N	Total documents number
D(t)	No. of documents with term t.

List of Abbreviations

Short Form	Abbreviations
Bi-LSTM	Bidirectional Long Short-Term Memory
BoW	Bag-of-Words
CNN	Convolutional Neural Network
FN	False Negative
FP	False Positive
GloVe	Global Vectors
GRU	Gated Recurrent Unit
LR	Logistic Regression
LSTM	Long Short-Term Memory
ReLU	Rectified Linear Unit
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency
TN	True Negative
TP	True Positive
2-L	2 Layer

Abstract

The world has recently seen unprecedented growth of social networks that have enabled internet users to share and communicate their views in various aspects of life. Analysis of opinion is one type of text mining that aims to discover human views and emotions about any particular event or topic. In this research work, we try to concentrate on comparing the performances of traditional machine learning models and deep learning models. Two widely used feature extraction techniques (BoW and TF-IDF) and SVM, LR and ensemble models are used for binary opinion classification. Various connections of CNN, GRU and LSTM are applied for hotel review dataset. We also compare our result by using two-word embeddings namely GloVe and FastText. In terms of accuracy, the CNN based networks perform well and 2-L CNN + GRU accomplishes the highest accuracy of 87.3%. Furthermore, the highest accuracy for traditional learning is achieved by the bagging ensemble, which is about 84.7%. In my work, Skip-Gram and combinations of different N-gram have not been considered and BoW or TF-IDF to dense document vector has not been applied.

Chapter 1

Introduction

1.1 Introduction

The world is going to be digitalized day by day and everything is going to be online based. Nowadays, people can express their opinions and feelings through online based media. So, for the business or many other purposes we need to analyze these opinions. For this analyzing purpose we need sentiment analysis. Sentiment analysis mainly relates to contextual text mining which measures or extracts relative and useful information from a text which helps an organization to understand their brand's value. Sentiment analysis gives an organization the basic concept about their products what the customer really cares about. Day by day online business is becoming more and more popular and a lot of reviews are created every day. So, it's a very first growing field and deep learning models makes it very effective to determine a customer's attitudes and feelings towards a particular product. Sentiment analysis with different kinds of deep learning models is very useful and give us accurate results towards determining opinions.

1.2 Background Study

A. Bag of Words (BoW): BoW aims to extract valuable features from text so that we can apply machine learning algorithms on it as we cannot feed our raw text data directly into the learning algorithms. BoW just keep track a count of occurrences of words and this process is done by converting the raw text into bag of words. To create a Bag-of-Words model first of all, we need to assemble the whole corpus of documents then make a list of all unique words from the corpus and at last revolve the unique words into vectors which means scoring the word. When a new sentence comes in the corpus and it contains new words then the vocabulary size will be increased, and length of the related vectors will also be increased [1].

B. Term Frequency-Inverse Document Frequency (TF-IDF): TF-IDF mainly focuses on getting the documents with similar words. TF-IDF is the measurement of how relevant a word or the originality of a word to a document is compared with a collection of documents. The term frequency is the number of times a term appears in a document and the inverse document frequency is the measure of how common a word in the overall set of documents [2]. By multiplying these two-frequency metrics we get the desired TF-IDF. TF-IDF has two parts namely term frequency and the inverse document frequency. The Term Frequency is defined as:

$$TF(term) = \frac{n(t)}{N(t)} \tag{1.1}$$

Where, n(t)= No. of times the term t arises in the document and N(t)= Total no. of terms

Again, the IDF is expressed as:

$$IDF(term) = \ln \frac{N}{D(t)} \tag{1.2}$$

Where, N= Total documents number and D(t)= No. of documents with term t.

C. Convolutional Neural Network (CNN): CNN is basically performed the task of computer vision like image processing or classification and a filter or sliding window is used to process the image. But in the era of Natural Language Processing, nowadays CNN model is frequently used in the text classification and sentiment analysis. A CNN consists normally of many layers of convolution but includes other components as well. The final classification layer of a CNN is the output of the last convolution layer. In the field of image processing, the corresponding pixels are used as the input and in the field of NLP text analysis, matrix of the corresponding sentences are used as input and in the matrix each row presents a vector. And in text classification, these vectors are called the word embeddings [3].

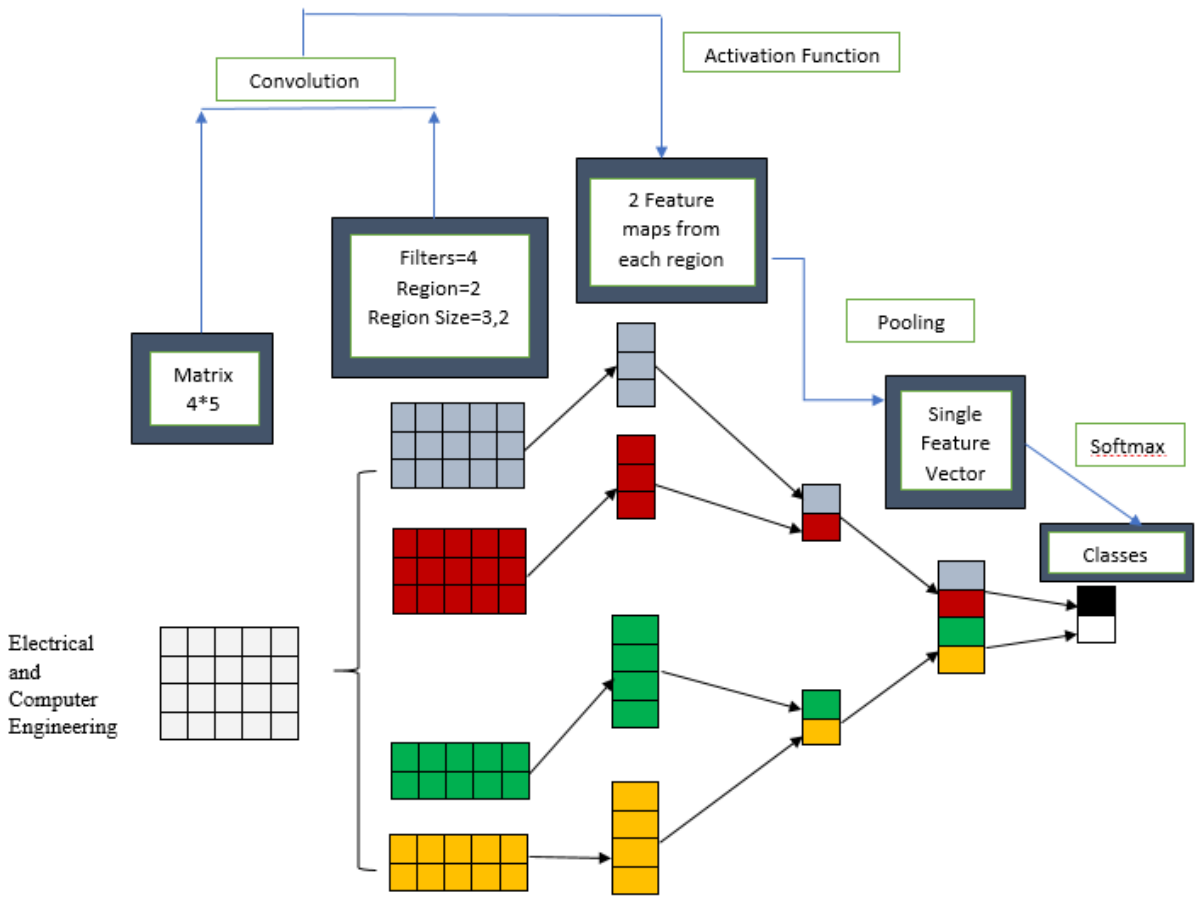

Fig. 1.1: CNN Network for NLP

In the above figure, we try to describe how the Convolution network works. In this figure, there is a matrix of 4*5 shape which is the input of the model. There are 4 filters, and those filters are divided into two region and the region size is 3 and 2. 1-max pooling is as pooled layer and softmax activation is used as activation function.

D. Pooling: Down sampling and pooling are basically same kind of things. The aim of the pooling layers is to yield spatial variance. In accordance with the pooling window, it reduces the size of the matrix.

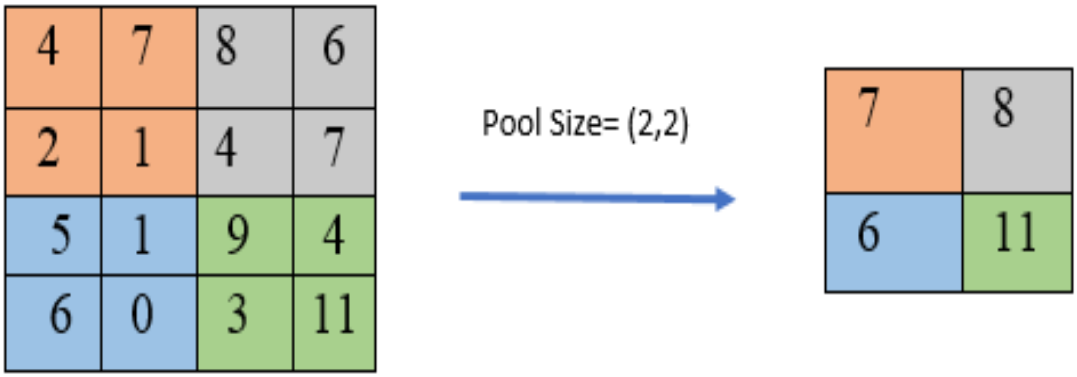

Fig. 1.2: Max Pooling

E. Long Short-Term Memory (LSTM): In the traditional Recurrent Neural Network (RNN), every output is dependent on the previous. So, a dependency creates, and every parameter need to remember the previous one. There is a cell in the RNN network which remember all the previous information that has been calculated. Recurrent neural networks leverage backpropagation through time (BPTT) algorithm to determine the gradients, which is slightly different from traditional backpropagation as it is specific to sequence data. One of RNN's appeals is to allow past knowledge to be connected to the present mission, such as using previous video images, to remind the interpretation of the current picture. The LSTM is capable of removing or inserting information in the cell status, regulated by mechanisms known as gates. But in case of long-term secular dependencies, the traditional RNN does not perform well enough due to the vanishing gradient error problem. So, the term LSTM come into account. In the traditional RNN, there is only the hidden layers but in the LSTM, it has both hidden layers and cell states. If a '0' for a given value is released in the cell state the forgotten gate needs the status of the cell to fully forget this piece of data. A '1' also means that this whole piece of knowledge is recalled by the forgotten gate. This contribution is compounded to cell status by the sigmoid function. The main advantage of having the cell state is that it has a set of gates which remove or add information in the system. Forget gate, Input Gate and Output gate are some gates that are used in the LSTM model [4].

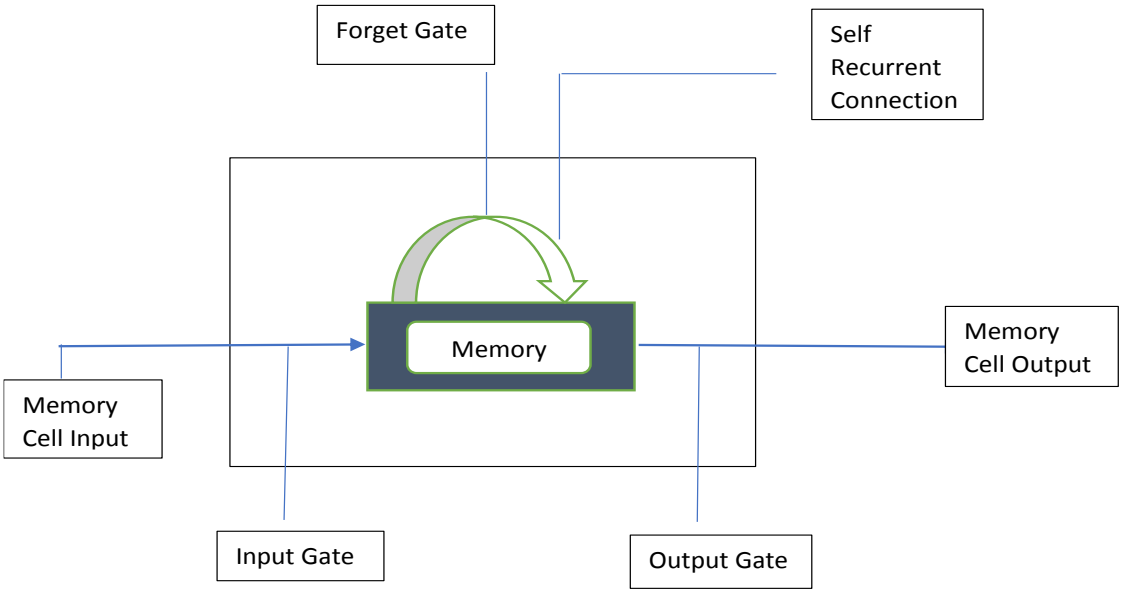

Fig. 1.3: LSTM Memory Cell

F. Gated Recurrent Unit (GRU): GRU has a extra set of gates for controlling the flow of information between different layers. Basically, GRU is much simpler than the LSTM model because it does not have separate memory cell and it has fewer gates than LSTM. As it does not have separate memory unit, it requires less time to train the model. GRU has two gates namely reset and update gate [5].

1.3 Literature Review

Goularas, Dionysis, and Sani Kamis [6], used three different data sets i.e., SemEval2014 Task9-SubTask B, SemEval2016 full data Task4 and SemEval2017 development data consisting of total 32.000 tweets. They used different kinds of deep learning models and compared those results. They show how the accuracy varies with respect to different combination of convolutional neural networks (CNN) and recurrent neural networks (RNN). They also used GloVe and Word2Vec embedding to increase the performance of the models and the performance with GloVe embedding is higher than the Word2Vec embedding. The accuracy is

found about 49% when they used Single CNN network with Word2Vec embedding and accuracy is about 51% when they used Single LSTM network with Word2Vec embedding. The highest accuracy is about 52% in the Word2Vec embedding when they used Multiple CNN's and bi-LSTM Networks. In the GloVe embedding, the accuracy is found about 54% when they used Single CNN network, and accuracy is about 55% when they used Single LSTM network. The highest accuracy is about 59% in the GloVe embedding when they used Multiple CNN's and bi-LSTM Networks. The authors also experimented with different combinations of CNN and RNN such as Single 3-Layer CNN and LSTM Networks, Multiple CNN's and LSTM Network, Single 3-Layer CNN and bi-LSTM Networks.

Ombabi, W. Ouarda and A. Alimi [7], in order to preserve long-term dependencies, they suggested a new deep learning model for arabic sentiment analyzes based on a single layer of CNN architecture for extraction of local features and on two layers of LSTM. To obtain the ultimate classification the feature maps learned by CNN and LSTM are moved to the SVM classification. The FastText word embedding model follows this model. Extensive studies on a multi-domain corpus show that this model performs in an excellent rating of 90.75%. In addition, multiple embedding structures and classifiers are used for the proposed model. The findings revealed that SVM is the most effective classifier with an accuracy of up to +3.92%. The proposed model produced promising performance and superior state-of-the-art approaches on many benchmarks, with an increase of up to 11.6% in accuracy, since the CNN was successful in removing features and repeating LSTM.

P. Ray eta A. Chakrabarti [8], proposed a deep learning approach model for aspect extraction from different texts and analyze the sentiment corresponding to the aspect for improving the model performance. They used a 7-layer CNN network in each opinionated sentence to tag each aspect. In this research paper, movie reviews from SST-1 and Restaurant reviews from SemEval Task 4 is used which contains more than 500,000 reviews. The overall accuracy for CNN+ Rule-based approach is about 0.75 and accuracy is 0.80 with CNN. The most improved approach is CoreNLP + Rule-based for which the accuracy is about 0.87. It also demonstrated here that CNN networks perform better than traditional methods.

Siwei Lai, Liheng Xu, Kang Liu and Jun Zhao [9], demonstrated that, the traditional methods for sentiment analysis such as BoW + LR, Bigram + LR, BoW + SVM, Bigram + SVM has the lower performance rate than the deep learning models. They proposed a Recurrent Convolutional Neural Network (RCNN) to improve the model accuracy. They performed their experiments using the following four datasets: 20Newsgroups, Sentiment Treebank, ACL Anthology Network and Fudan Set. They compare the results of the traditional models with the improved deep learning models. For 20News dataset, the accuracy is about 92.81% when they apply BoW + LR method and the accuracy is 93.12% using the Bigram + LR method, the accuracy is 92.43% with BoW + SVM. On the other hand, the accuracy is about 94.79% by applying CNN model and it is about 96.49% after applying the RCNN model. A considering number of changes is observed here. Similarly, for all other data sets the deep learning-based models work more efficiently than the traditional methods.

In another research study, Mathieu Cliche [10], proposed a state-of-the-art Twitter sentiment classifier using the combination of CNN and Long Short-Term Memory (LSTMs) networks. Here SemEval-2017 Twitter dataset is used, and the embeddings are again fined tuned after using a subset of the unlabeled data to fine tune the embeddings. The author evaluated with three word embedding: the Word2Vec, GloVe and FastText and the GloVe embedding gives lower accuracy on the other hand Word2Vec gives the better performance. The accuracy for the following embeddings with the LSTM network is respectively 0.636, 0.630 and 0.633 for the 2016 test set which contains 20,632 tweets. For CNN (convolution size= [3,4,5], the accuracy is 0.643,0.637 and 0.640 respectively. At last, the author used an ensemble of 10 CNNs and 10 LSTMs which gives the accuracy of 0.648.

A research paper authored by Xuanxin Liu, Fu Xu, Yu Sun, Haiyan Zhang and Zhibo Chen [11], showed the effect of InceptionV3, ResNet50, Xception and Mobile Net CNN with three others different RNN units: simple RNN, LSTM and GRU on the10-fold test accuracy. They used Flavia dataset for leaves observation. When they experimented with ResNet50 and simple RNN, they got a mean accuracy of 0.7987 and experimented with the InceptionV3 and simple

RNN has 2,249,880 trainable parameters which gives mean accuracy of 1.000. Similarly, with Xception and MobileNet they achieved mean accuracy of 0.9989 and 0.9995 respectively. For LSTM, the combination of LSTM with MobileNet gives the highest mean accuracy and for GRU, the combination of GRU with Xception and MobileNet gives the highest accuracy. They proposed the C-RNN models for observation-centered plant identification and here the RNN units integrate features and CNN units extract features. The combination of MobileNet and GRU gives the best classification accuracy and computational overhead on the Flavia dataset.

Haotian Xu et al. [12] used the dataset of MEDLINE citations from November 2012 to February 2013 which contains 143,853 citations in total, from which 94,942 citations are considered for training. They proposed a word embedding Topic based Skip-gram model, which shows relationship between word semantic and topic models such as Latent Dirichlet Allocation (LDA), and in the end integrate it into distributed word embedding. They conducted their experiment using the combination of multimodal CNNs and the proposed topic-based model. They compared the Topic-based Skip-gram methods with several CNN-based models like CNN-ran, CNN-gn, CNN-glove, CNN-local and CNN-topic. In this paper, CNN-concat gives higher results than CNN-channel for 15 terms among 29 terms and CNN-concat performs better than CNN-channel for more balanced MeSH terms. It also shows that the performance of F1 score is much higher in CNN-based models than the traditional machine learning algorithms.

Again, in another paper, Shewalkar, D. Nyavanandi and S. Ludwig, [13] evaluate different kinds of Deep Neural Networks (DNN) for pesian speech recognition. The authors carried out their experiment by using the FARSDAST dataset. They use Fully Connected (FC) structure with three types of neuron models: sigmoid, ReLU and Maxout. The maximum accuracy is found when they experimented with Maxout model with 3 hidden layers and the accuracy is about 88.34%, on the other hand the minimum accuracy is found with sigmoid model and the accuracy is about 82.86%. Later phase in the experiment they used a Convolutional Maxout Deep Neural Networks (CMDNN) model to evaluate the performance of the system and eventually an accuracy of 89.42% which is higher than the before.

Meng Cai and Jia Liu [14], suggested that Maxout neurons are very much promising alternatives to sigmoid neurons. In their paper, they combine the two popular DNN structures, namely the convolutional neural network (CNN) and the long short-term memory (LSTM) recurrent neural network (RNN) with Maxout neurons. They contracted their experiments on the benchmark data sets released in the IARPA Babel Program. Their proposed models give 2.5–6.0% more accuracy over their corresponding CNN or LSTM RNN baselines across six language collections. After the combination of the system these results are taken. CMNNs give more TER reduction than the CNNs and the value is in between 4.3-6.00%.

N. Dang, M. Moreno-García and F. De la Prieta [15], used deep learning to overcome feeling interpretation problems such as sentiment polarity. This study has been published. A variety of data sets have been fitted with templates using a frequency-inverse frequency (TF-IDF) concept and a word embedding. Finally, a comparative study was conducted for the different models and input functions. Following an analysis of 32 papers, the most common models for the analysis of emotion polarity were defined as DNN, CNN and mixed approaches. Another point taken from the study was the fact that standard strategies such as CNN, RNN, and LSTM are independently evaluated on various datasets in those studies. Moreover, the findings presented in most articles are accurate without taking estimation time into consideration.

1.4 Conclusion

Sentiment analysis mainly focuses on extracting effective and useful information from any given review in the form of positive, negative or neutral. In the era of globalization, restaurant and hotel both are very much important and most frequently used. Nowadays, online review has become more and more popular because people generally use social media and other online sources to express their opinion towards a particular thing. For hotel, this kind of reviews are way more important and very much helpful for tourist to make efficient decision. In this paper, we mainly focus on different kinds of feature extraction techniques and deep learning approaches and how they work in different kinds of classification algorithms including ensemble.

Chapter 2

Methodology

2.1 Introduction

Our total experimental methodology is depicted in Fig.4 and Fig.5. First figure shows our methodology for traditional machine learning algorithms which includes some major pre-processing steps and converting the text into numbers using two very popular feature extraction methods, namely BoW and TF-IDF. Then two traditional models i.e., SVM and LR and two ensemble methods, namely Bagging and Voting have been employed and polarity has been identified for the review dataset. In the second figure, deep learning approach for text classification is shown which also included major pre-processing steps and after tokenization, padding and some other steps, the word embeddings are obtained. Then the dense embedding is passed through hidden layers and the output of the hidden layers is passed through the activation function and in the end, we get our desired output.

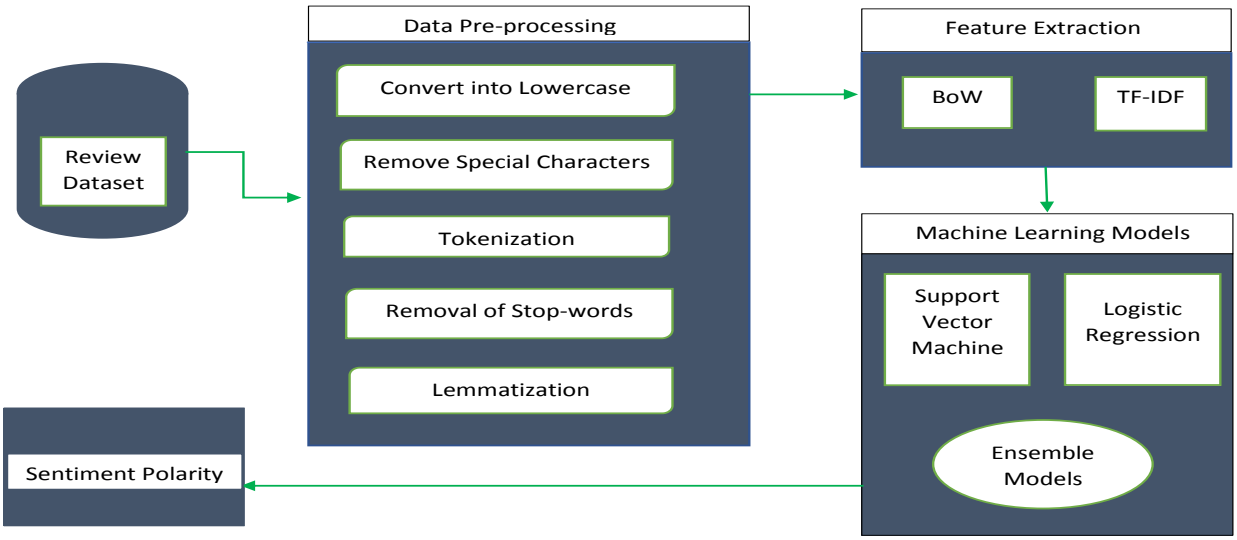

Fig. 2.1: Traditional Machine Learning Approach for sentiment analysis

Instead, terms are depicted by distance and angle in an embedding in which a vector displays the word's projection into a continuous vector space. The location of a word within the space of a vector is learned by the text and by the terms surrounding the word. After passing the hidden and activation layer our two-dimensional output is obtained.

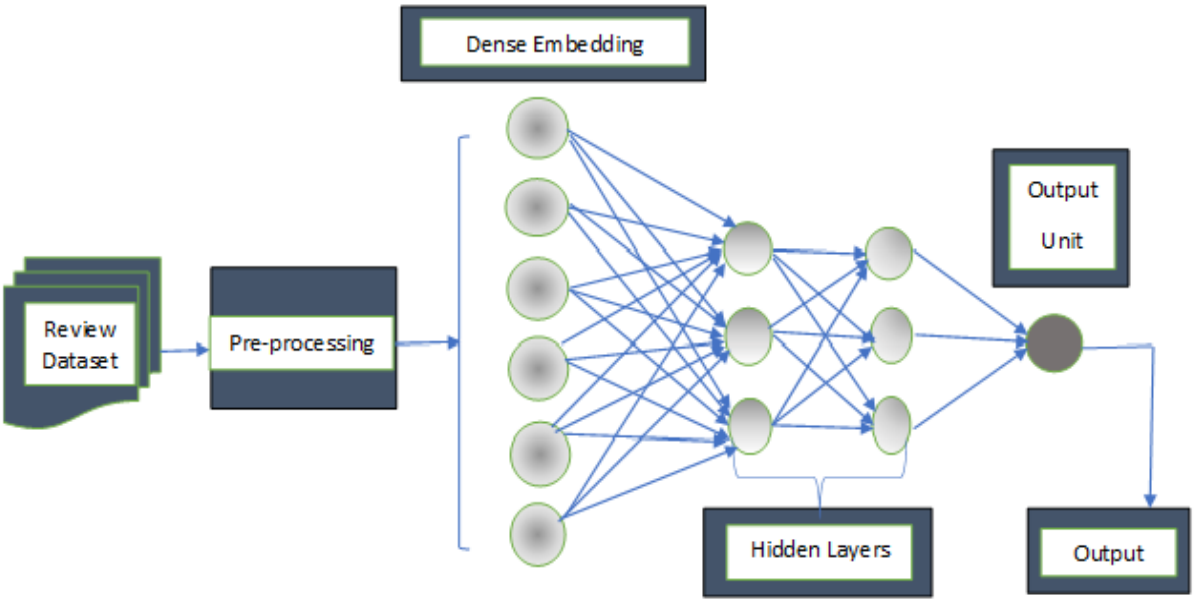

Fig. 2.2: Deep Learning Approach for sentiment analysis [16]

2.2 Dataset

Our experiment is evaluated on a subset of "515k Hotel Reviews Data in Europe". The original dataset contains 515,000 customer reviews from 1493 different hotels. For our experimental setup, we took only 15,000 reviews and 7,500 positive and 7,500 negative reviews had been considered to make the dataset balanced. The original dataset contains 17 columns, but in our dataset, we took only two columns because these two columns are enough to make our prediction [17].

2.3 Pre-processing

The pre-processing of text is a compulsory step in the production of natural text (NLP). In this pre-processing step, we need cleaning this kind of noisy text data until it can be incorporated into the machine learning model in a number of words of correct spelling, short words, special symbols, emojis, etc. The major pre-processing steps are given below:

1. **Removal of punctuation, number, and special character:** Special characters normally contain any character that including punctuation and whitespace, is not a letter or number. A series that contains only letters and numbers refers to special characters being omitted from a string**.** We need to remove the punctuation and other special characters from the string which are not important to make sentiment classification decision. So, in the preprocessing steps, we removed all the punctuation, numbers and useless characters for better performance.

2. **Converting to lowercase:** In this preprocessing step, we convert all the letters into lowercase because it helps in parsing and it does not affect the efficiency of the model.

3. **Tokenization:** Tokenization is a fundamental step in preprocessing. Tokenization is essentially breaking a raw text, or an entire raw text document into smaller chunks and each of these smaller chunks are called tokens. We frequently used this tokenization method to easily interpret and analyze the text. We used this tokenized form to analyze the number of words present in the whole text and the frequency of the word.

4. **Removal of stop-words:** Stop-words are those most common words in the language which does not provide much value or meaning to a sentence. As a result of removing the stop-words, we need less time to train the model and potentially improve the performance. The NLTK library is one of the most widely used Python libraries. Stop word elimination is enabled by NLTK, and there is a list of stop words in the corpus module. We can cut the text into words to remove stop words from a sentence, and the word is removed if it exits the NLTK list of stop words.

5**. Lemmatization:** Lemmatization refers to convert the inflectional and derivationally related forms of a word into its root form of word. In this lemmatization process, the root word is called a lemma and this process is more accurate than stemming (non- dictionary-based process). Lemmatization is a dictionary-based process and gives the dictionary word meaning when we convert it into common form [18].

2.4 Word Cloud Representation

Tag clouds or word clouds are visual depictions of word frequency that make words in a source text more prominent. The bigger the word in the illustration, the more regular the word in the text was. By recognizing terms that also appear in a series of interviews, records, or other text, this form of visualization may aid evaluators with exploratory textual analysis. A word cloud is a series of words of various sizes. Word clouds are a cost-effective and convenient way to view text results. One of the difficulties of reading word clouds is to stress the frequency and not actually the meaning of terms. There are numerous word and tag cloud generators available freely on the internet and the process is simple. Evaluators can literally import text into a text box (for example, a series of interviews) and a graphical representation of words is created by this method. The bigger and bolder the expression, the more frequently it is referred to and the more relevant it is inside a particular document. Tag clouds are easy-to-use and affordable way to display text results. One of the difficulties of simplifying word clouds is that the display show stresses word frequency, not generally its value. If strikingly different terms are used with the same definition, tag clouds cannot correctly represent the text information. They also do not have context, so the meaning of particular words will be lost. Owing to these limits, word clouds are ideally suited to qualitative exploratory studies [19].

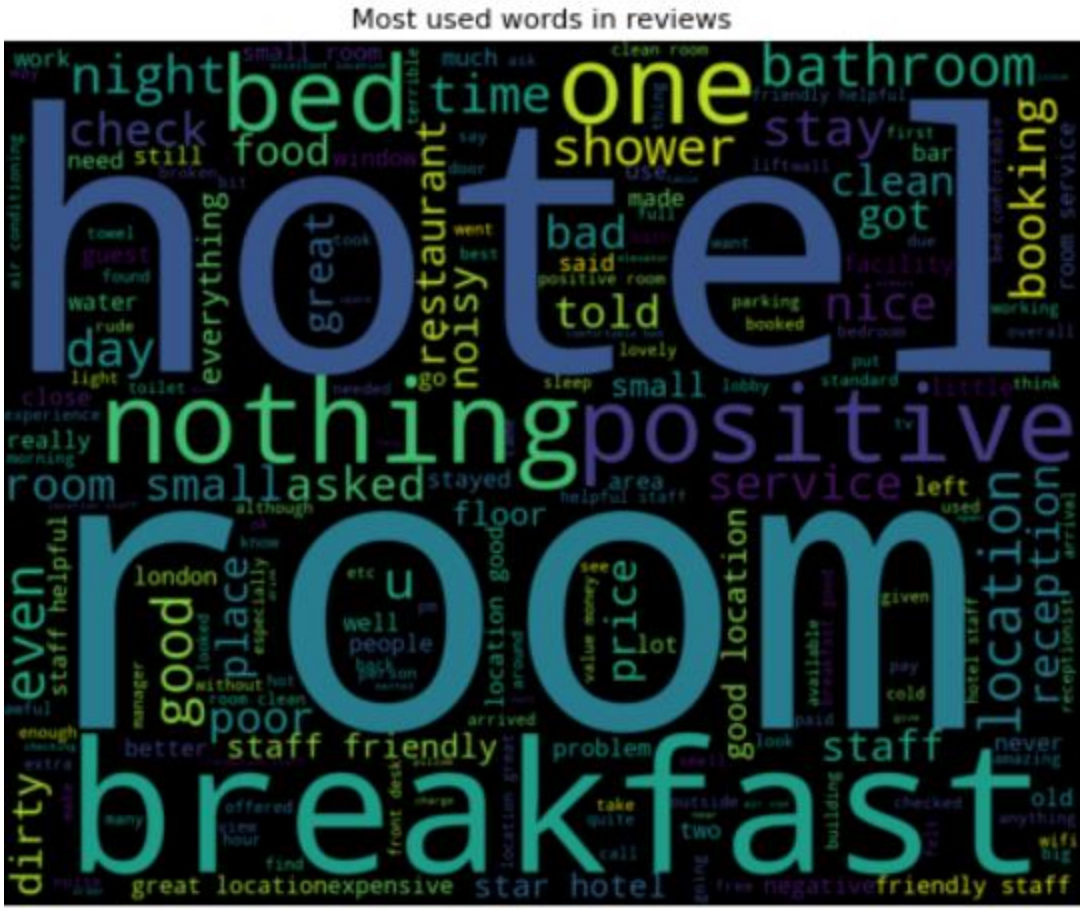

Fig. 2.3 : Word Cloud Representation of Hotel Dataset

2.5 Classifiers

2.5.1 Support Vector Machine (SVM): SVM is a supervised and associated learning algorithm that recognize pattern and analyze the data. Each data item is compiled as a point by SVM algorithm. The plotting takes place in an n-dimensional field where n is the number of characteristics of a given data. Then the most adequate hyperplane may be graded, easily separating the two or more groups. SVM performs best when it prefers the line that divides the details that is as far removed as possible from the data points in the wardrobe. There are some which are nearest to the hyperplane and the data points are helpful to define separating line. The

hyperplane is the preference of the data points and the hyperplane dimension varies with the number of input functionalities. Margin is the gap between two data lines on the nearest data points of different classes. The difference between the line and the supporting vectors is perpendicular. However, it is generally used for problems of grouping. In the SVM algorithm, we mark each data object with the value of a particular coordain in n-dimensional space. We then classify by discovering the hyperplane that makes very good distinctions between the two groups [20].

2.5.2 Logistic Regression (LR): Linear regression is not bounded and not useful for classification problem because it needs only the continuous data. Because of the advantage of using the categorical or continuous data, and this brings logistic regression into frame. LR is a logical Sigmoid function algorithm used to transform its output, thus returning a discrete probability value. As a result of using the sigmoid function its value is bounded between 0 and 1. There is a judgment limit that positions the probabilities in the positive and negative classes. In the LR model, maximum likelihood estimation (MLE) method is used to determine the best fit line for coefficients and the procedure is repeated until Log Likelihood (LL) does not change significantly. LR uses the odds ratio. A logistic regression is not necessarily analyzed the odds, but does the logarithmic transformation of the odds ratio, the log odds and as it does logarithmic transformation of the odds ratio, output is not always straightforward [21].

2.5.3 Voting Ensemble: By integrating the forecasts from many models, a voting ensemble functions. The hard voting approach uses forecast laboratories and consensus laws, while the soft voting method forecasts the mark on the grounds that the sum of the expected probabilities is argmax/largest predicted. It may be used when classifying or regressing. For classification, the predictions are summarized for each label for each grouping and the label is supposed to be accepted by plurality vote. The voting group is constrained to treating each model the same way, which ensures that each model contributes to the forecast equally. This is an issue if some models are good in some cases and some are bad in others [22].

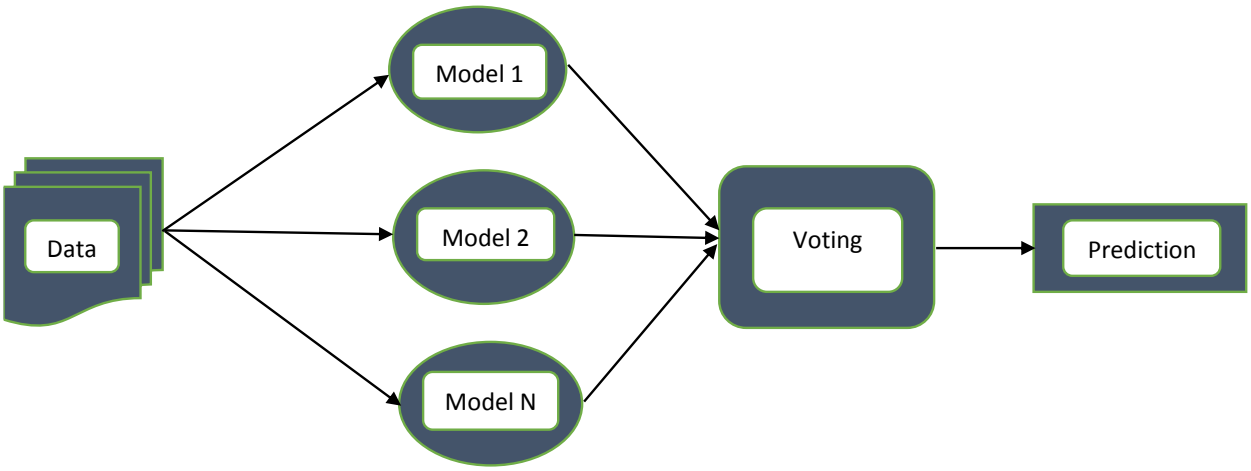

Fig. 2.4: Voting Ensemble

2.5.4 Bagging Ensemble: The grouping of several iterations of a predicted model is bagging, also known as bootstrap aggregating. Each model is independently trained, using an averaging method, and combined. Bagging is used to reduce the uncertainty in the description of a decision tree. The primary goal of bagging is to produce less variation individually than every model has. Finally, the average of the outputs expected by the individual students in the case of regression. In order to classify the class that is most voted (hard votes), the highest average probability is considered to be the result (soft voting). Combining approaches and merge multiple decision-taking classifiers to achieve improved predictive efficiency than a single decision-tab classifier. A community of slow learners assemble to form a strong learner, improving the model precision, is the core concept behind the ensemble model. As the final forecast is based on average projections from sub-set trees, the classification and regression model can not be exactly categorized. It correlates to the simple learners of the initial dataset for any random subset (bootstrapping). Due to the parallel ensemble, meaning each model can inherit somewhat different characteristics [23].

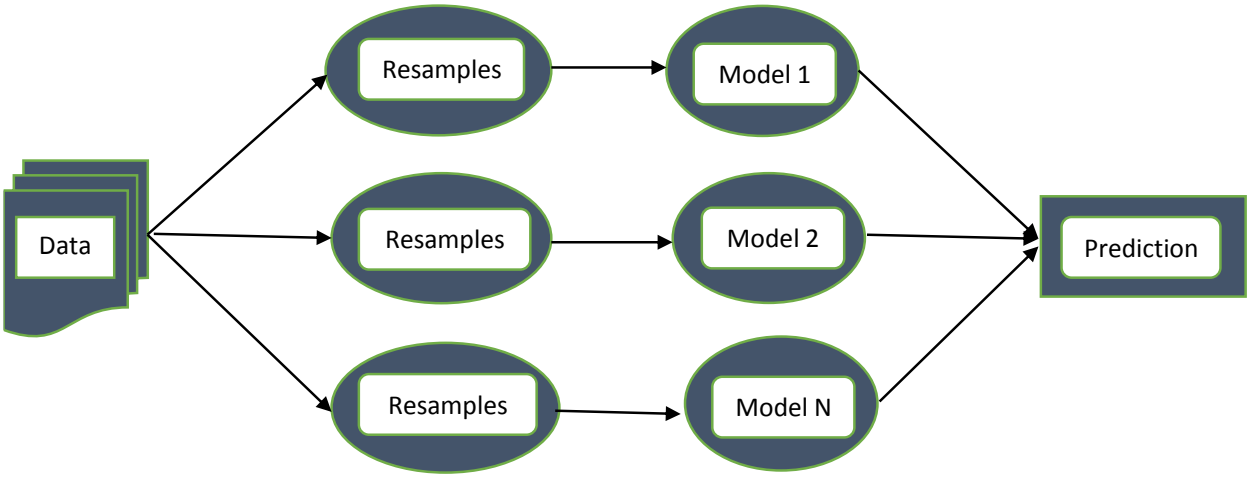

Fig. 2.5: Bagging Ensemble

2.6 Word Embedding

Word embedding is a method of representing the words as a dense vector. It has some major advantages over the traditional bag-of-words method. In the traditional model, the words are represented as a huge sparse matrix but in embedding it is replaced by the dense vector. The suggested technique will improve traditional word embedding and outperform the emotion embedding suggested previously in the Stanford Sentiment Treebank, for both really well and binary grades. In vector space, the location of a word is learned from the text and is dependent on the terms surrounding the word. In the learned vector space, the location of a term is called its embedding [24].

Keras has a built-in layer that can be used on text data on neural networks. It requires an integer encryption of the input data, such that any term is represented by a single integer. Here, first of all, the layers are initialized with weights which are randomly picked and then all the words will learn an embedding. Actually, the model is learnt from the scratch and no pre-trained embedding is required. This layer is the first layer in the model which has 3 arguments namely

the input_dim or the vocabulary size, the output_dim or the size of the vector space and the input length. In our experiment, the vocabulary size is 15,000, the output dim is 100 and the max length is 350.

Besides this keras embedding layer, we also use two pre-trained embedding namely GloVe and FastText embedding. The GloVe or GloVe global vectors is an expansion of the word2vec approach developed by Pennington et al. in Stanford for the efficiency of word vector. Unlike word2vec, GloVe tries to render specifically what word2vec indirectly does. GloVe integrates in word2vec data on world-wide matrix factoring, such as LSA, and local context-based learning. GloVe employs figures in the whole text corpus rather than the use of a window to define a local context to create an abstract word context or word co-occurrence matrix. The effect is a paradigm of learning that can usually lead to stronger word integrations. In our experiment we use "glove.6B.300d" which contains 6B tokens, 400k vocabulary and 300d vectors.

FastText puts together some of the best ideas introduced in the last decades by the linguistic analysis groups. This involve describing terms like bag-of-words and bag of n-gram expressions, as well as using sub words and exchanging knowledge through classes through a secret monitor. They also use a hierarchical softmax to boost the estimation using the unbalanced class distribution [25]. The vector representation for words not in the dictionary (OOV words) can be shown as they can also be separated in character n-grams. Both words2vec and glove do not provide any representations of the vectors for words not in the dictionary. FastText also describes the text by a small vector that is obtained by summing vectors that fit the terms in the text. FastText embedding takes advantage of subword knowledge to create word embedding. Interpretations of n-grams are acquired and words as n-gram vectors are interpreted. In fastText, any word of the vocabulary has a tiny vector. It is spread across all categories, allowing information on vocabulary that have been acquired with one category to be used for other categories. In this experimental setup we use "Wiki-news-300d-1M.vec" which contains 16B tokens and 1-million-word vectors.

2.7 Neural Networks

Our experimented neural network framework is mainly based on CNN, LSTM and GRU networks for the assessment of hotels reviews.

• **Single CNN Network:** In this experiment we use a single CNN network. In this network, there is Conv1D layer, GlobalMaxPooling1D layer, Flatten layer, Dropout layer and Dense layer. The sentence vector is elaborated with a Conv1D layer in which 64 filters are used and the kernel size is 1×3. GlobalMaxPooling1D has a pool size of 2. In this experiment, we used 50% dropout. The output layer is 2 dimensional because of our positive and negative reviews.

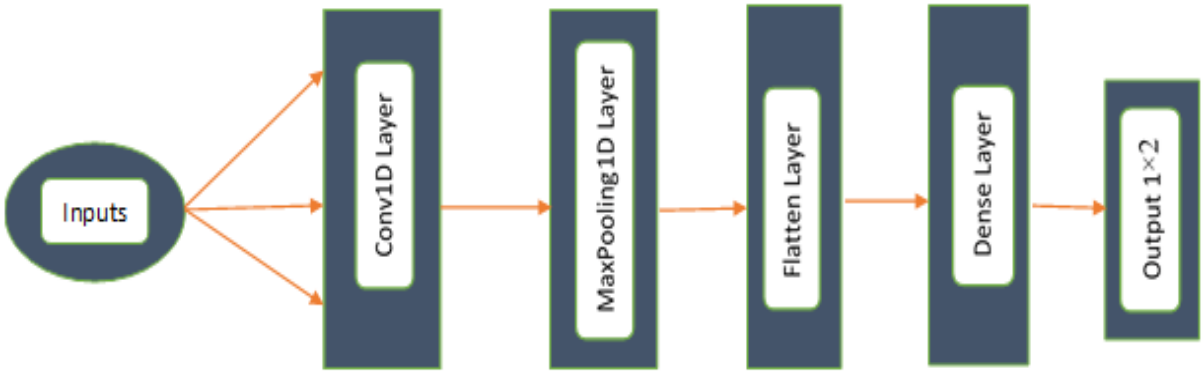

Fig. 2.6: Single CNN Network

• **Single Bidirectional-LSTM Network:** In this configaration, there is LSTM layer, Dropout layer and Dense layer. In the LSTM layer there is 64 hidden units or latent dimension, and recurrent dropout is 40%. ReLU activation is used in the dense layers except the output layer. And again, the output layer is 2 dimensional.

• **Single GRU Network:** We also experimented with single GRU network in which there is a bidirectional GRU layer and the number of neurons or units in this GRU layer is 128. There are also dropout and dense layers. The dropout is 50% here. Sigmoid activation function is used to obtain the output and the output is 2 dimensional due to the polarity.

• **2-layer CNN and Bi-LSTM Networks:** This configuration is illustrated with a 2-layer CNN and a LSTM networks. In this set of experiment, the input is passing through a 2-layer CNN networks in which the kernel size is 1×3. Then the output of the CNN models is given to the LSTM network. Dense and Flatten layers are followed by the LSTM layer and final output is obtained by using the sigmoid activation function.

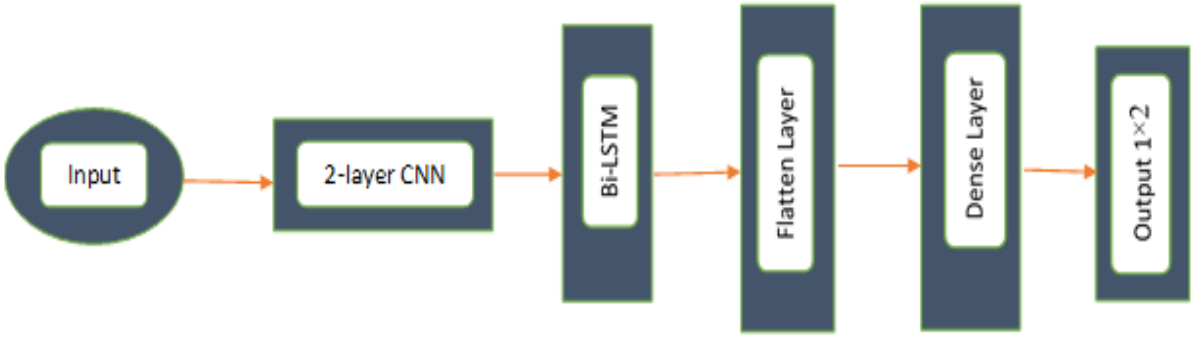

Fig. 2.7: 2-L CNN and Bi-LSTM Network

• **2-layer CNN and GRU Networks:** This setup is demonstrated with a 2-layer CNN network and an GRU network. The input passes through a 2-layer CNN network in this series of tests, in which the kernel size is 1-3. The GRU layer is accompanied by dense and flatten layers, and final output is obtained by using the sigmoid activation function which is 2 dimensional.

• **Single CNN and LSTM Networks:** This setup is almost similar to the 2-layer CNN and Bi-LSTM. But in this case, we used the LSTM layer instead of Bi-LSTM and in place of 2-layer CNN we use single CNN. This is a ensemble model which is the combination of CNN and LSTM network to know how the model works on ensemble model. Normally, most of the cases it will improve the performance of the model. In my work I try many combinations of CNN and LSTM to compare the result. In my work this model gives the highest performance when applied with the FastText embedding. The word embedding improved the performance of the model.

2.8 Model Hyper-parameters

Different parameters and configurations have been evaluated empirically. The CNN configurations were set to 0.5 before the recurrent layer to minimize overfitting for the adopted convolutional layer used by various width fitters, ReLUs as the activation feature, and padding was set to zero. The hidden state dimensionality was set to 128 and 64 for LSTM configurations and the sigmoid function was used as an activation function. In architecture, the total number of epochs (4-10) was set.

2.9 Evaluation Metric

In this experiment, we considered Accuracy, Precision, Recall and F-measure (F1) as our evaluation parameter. We also considered the RMSE error to compare different kinds of models. The formula of Accuracy is defined as:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{2.1}$$

Precision is expressed as:

$$Precision = \frac{TP}{TP+FP} \tag{2.2}$$

Similarly, Recall and F-Measure is defined as follows:

$$Recall = \frac{TP}{TP+FN} \tag{2.3}$$

$$F_Measure = \frac{2TP}{2TP+FP+FN} \tag{2.4}$$

where, TP: TP means true positive, which is demonstrated by the number of ratings categorized as positive and they actually belong to the positive class.

TN: TN means true negative, which is demonstrated by the number of ratings categorized as negative and they actually belong to the negative class.

FP: FP is false positive, which is demonstrated by the number of ratings that is graded as positive, but their real class is negative.

FN: FN is false negative, which is demonstrated by the number of ratings that is graded as negative, but their real class is positive.

2.10 Conclusion

In this chapter, we tried to express how these algorithms work on different kind of approaches. We generalized our proposed model and visualize it. We also expressed the formula that defines our evaluating metrics.

Chapter 3

Results and Discussion

3.1 Introduction

In this section, we discuss empirical outcome of our experiment and how different kinds of models works on this data set. We evaluated these models based on accuracy, precision, recall and F-1 score. We decorate our results in Table 3.1- Table 3.5 and Fig. 3.1- Fig. 3.5 according to the model.

3.2 Results

In terms of accuracy, the CNN based networks perform well and 2-L CNN + GRU accomplishes the highest accuracy of 87.3%. Furthermore, the highest accuracy for traditional learning is achieved by the bagging ensemble, which is about 84.7%. It is verified from the findings that strong performance for hotel review is obtained by both function extraction strategies and classifiers. The noteworthy findings reveal that the voting ensemble model offers the best performance for the BoW and bagging ensemble for the TF-IDF.

3.2.1 Traditional Machine Learning Algorithms

3.2.1.1 BoW: Firstly, we demonstrated BoW technique for our hotel review dataset. It is verified that Voting Ensemble accomplishes the highest accuracy by obtaining 83.7% for hotel review dataset whereas Bagging Ensemble accomplishes the highest Recall, Precision and F1 Score by obtaining 88.1%, 82.2% and 85.0% respectively for the review dataset. Furthermore, it is inspected that SVM accomplishes the lowest accuracy by obtaining 81.0% for hotel review dataset. The output result is shown in Table 3.1 when we used bag-of-words feature extraction technique on several machine learning models. Fig. 3.1 depict graphical rendition of our model's performance.

Table 3.1: Performance of different classifiers by using BoW technique

Classifier	Accuracy	Recall	Precision	F1 Score
SVM	0.810	0.841	0.820	0.832
LR	0.830	0.852	0.810	0.831
Voting Ensemble	**0.837**	0.877	0.814	0.845
Bagging Ensemble	0.831	**0.881**	**0.822**	**0.850**

From the table 3.1 it is verified that accuracy is high for ensemble model compared to the other models. Table 3.1 depicts that bagging ensemble scoring the highest recall, precision and f-1 measure.

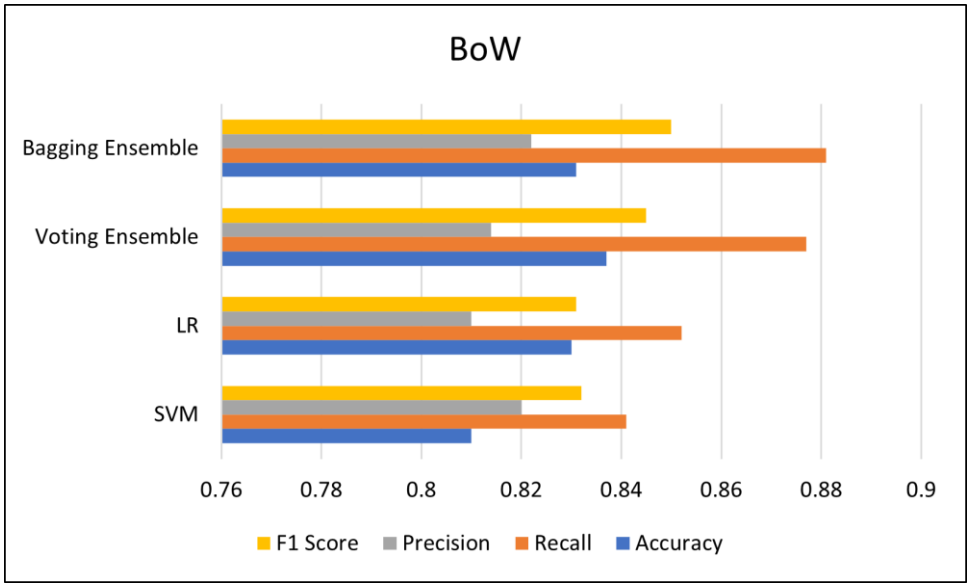

Fig. 3.1: Performance of different classifiers by using BoW technique

3.2.1.2 TF-IDF: Secondly, we examined TF-IDF technique for our hotel review dataset. It has been confirmed that for hotel review dataset, Bagging Ensemble has been able to obtain the highest accuracy by securing 84.7%.

Table 3.2: Performance of different classifiers by using TF-IDF technique

Classifier	Accuracy	Recall	Precision	F1 Score
SVM	0.822	0.857	0.805	0.830
LR	0.838	0.892	0.808	0.848
Voting Ensemble	0.840	0.896	**0.809**	0.850
Bagging Ensemble	**0.847**	**0.901**	0.802	**0.859**

Moreover, Voting Ensemble accomplishes the best precision with 80.9% score. The desired output result is shown in Table 3.2 when we used TF-IDF feature extraction technique on several machine learning models.

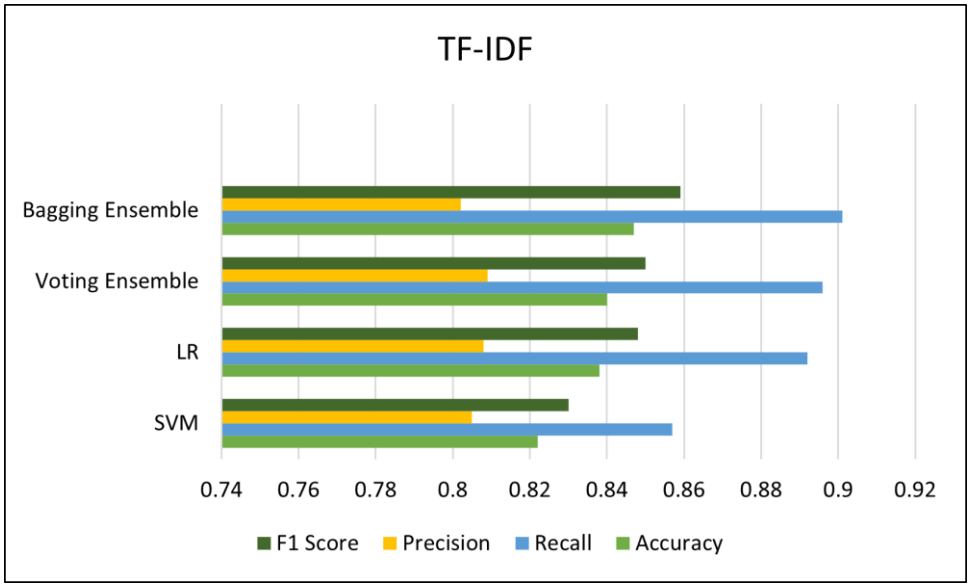

Fig. 3.2: Performance of different classifiers by using TF-IDF technique

Fig. 3.2 depict graphical rendition of our model's performance. Furthermore, it is observed that SVM accomplishes the lowest accuracy by obtaining 82.2% for hotel review dataset whereas Bagging Ensemble accomplishes the highest Recall, and F1 Score by obtaining 90.1%, 85.9% sequentially for the review dataset.

3.2.2 Neural Network Models

3.2.2.1 Without Pre-trained Embedding: In case of neural network, firstly, we demonstrated word embedding with no pre-trained vectors for our hotel review dataset. In our experiment, different layout of deep learning procedures based on CNN, LSTM and GRU networks were checked for sentiment analysis in hotel data. It is verified that Single CNN+ LSTM accomplishes the highest accuracy by getting 84.1% score for hotel review dataset whereas Single CNN+ LSTM accomplishes the highest Recall, and F1 Score by obtaining 89.2%, 85.5% sequentially for the review dataset. Furthermore, it is inspected that Single GRU accomplishes the lowest accuracy by obtaining 81.5% for hotel review dataset.

Table 3.3: Sentiment prediction with different layout of CNN, LSTM and GRU with no pre-trained embedding

Model Network	Accuracy	Recall	Precision	F1 Score
Single CNN	0.834	0.866	0.821	0.852
Single Bi-LSTM	0.829	0.860	0.805	0.831
Single GRU	0.815	0.851	0.812	0.829
2-L CNN+ Bi-LSTM	0.835	0.871	**0.824**	0.849
2-L CNN+ GRU	0.829	0.870	0.798	0.824
Single CNN+ LSTM	**0.841**	**0.892**	0.822	**0.855**

The experimented output result is shown in Table 3.3 when we used word embedding with no pre-trained vectors. Fig. 3.3 depict graphical rendition of our model's performance.

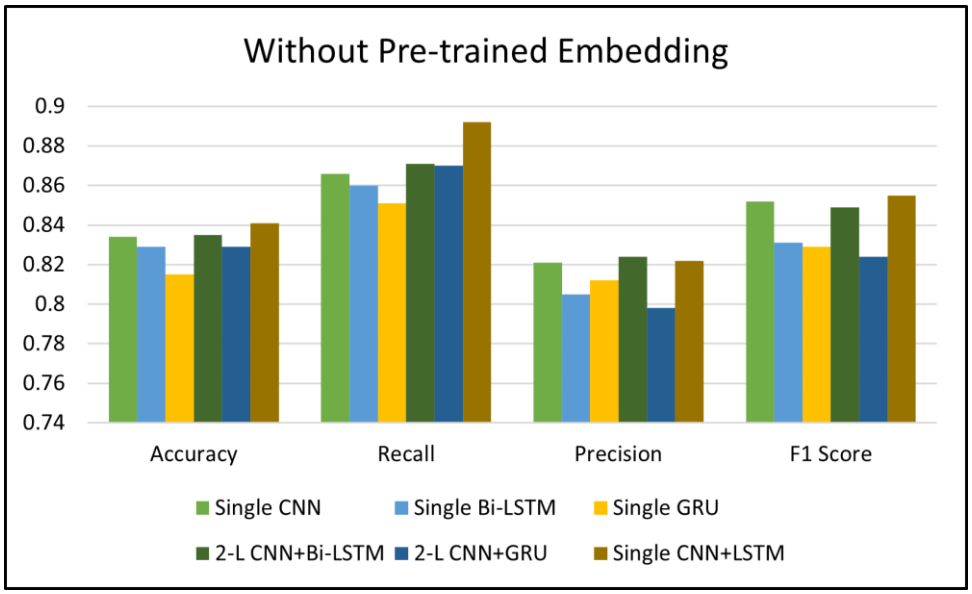

Fig. 3.3: Sentiment prediction with different layout of CNN, LSTM and GRU with no pre-trained embedding

3.2.2.2 GloVe: Secondly, we considered GloVe word embedding for our hotel review dataset. It has been verified that for hotel review dataset, the layout combination of single CNN and LSTM obtain the highest accuracy by securing 86.1% whereas Single GRU accomplishes the lowest accuracy with 83.2%. In case of precision and recall, Single CNN + LSTM gives the best result on the other hand, Single GRU gives the worst output. With the term F1, 2-L CNN + GRU accomplishes the highest recall by securing 93.0% for our experimented data set whereas Single GRU accomplishes the lowest recall with 86.1%. Single GRU also accomplishes the lowest precision and f-1 score by obtaining 81.0% and 83.9% respectively. From the table 3.4 it is also assumed that when we apply the ensemble model with the combination of different CNN and LSTM models, then the performance of the model increases with a certain amount.

Table 3.4: Sentiment prediction with different layout of CNN, LSTM and GRU with GloVe embedding

Model Network	Accuracy	Recall	Precision	F1 Score
Single CNN	0.851	0.910	0.852	0.881
Single Bi-LSTM	0.846	0.896	0.832	0.859
Single GRU	0.832	0.861	0.810	0.839
2-L CNN+ Bi-LSTM	0.854	0.921	0.844	0.894
2-L CNN+ GRU	0.858	**0.930**	0.835	0.889
Single CNN+ LSTM	**0.861**	0.920	**0.882**	**0.901**

The desired output result is displayed in Table 3.4 when we used GloVe word embedding for evaluating our results. Fig. 3.4 depict graphical rendition of our model's performance.

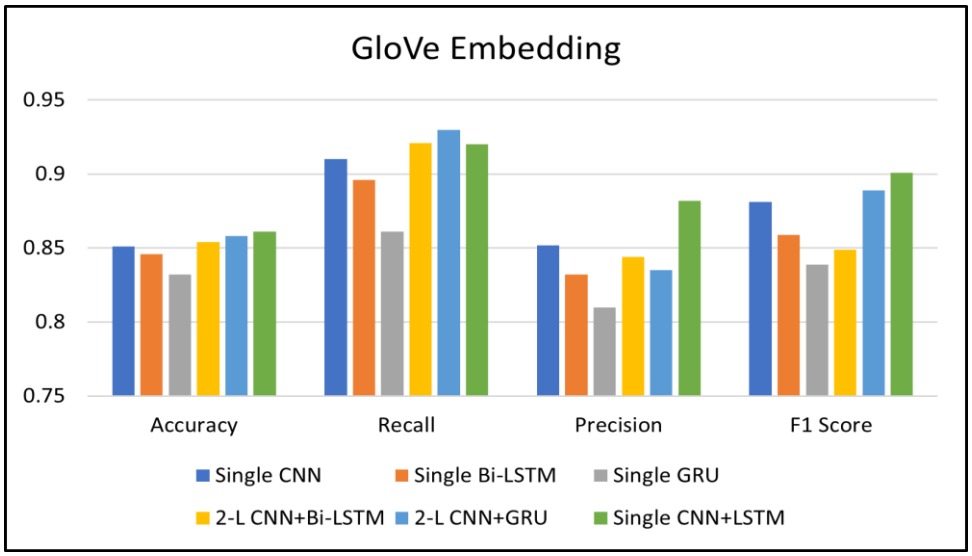

Fig. 3.4: Sentiment prediction with different layout of CNN, LSTM and GRU with GloVe embedding

3.2.2.3 FastText: This time, we considered FastText pre-trained word embedding for our hotel review dataset with different layout of CNN, LSTM and GRU networks. From Table 3.5, it has been confirmed that for our hotel review dataset, 2-L CNN + GRU has been able to obtain the highest accuracy by securing 87.3%. Moreover, Single GRU performs very badly and has been able to obtain 83.2% accuracy. Single GRU also accomplishes the lowest precision and f-1 score by obtaining 81.0% and 83.9% respectively. From the table 3.5 it is also assumed that when we apply the ensemble model with the combination of different CNN and LSTM models, then the performance of the model increases with a certain amount.

Table 3.5: Sentiment prediction with different layout of CNN, LSTM and GRU with FastText embedding

Model Network	Accuracy	Recall	Precision	F1 Score
Single CNN	0.841	0.900	0.823	0.861
Single Bi-LSTM	0.851	0.909	0.852	0.880
Single GRU	0.832	0.861	0.810	0.839
2-L CNN+ Bi-LSTM	0.849	0.899	0.842	0.872
2-L CNN+ GRU	**0.873**	**0.953**	**0.901**	**0.922**
Single CNN+ LSTM	0.864	0.929	0.884	0.905

From the table 3.5, we also claimed that 2-L CNN + LSTM also performs well in the recall, precision and F1 score and obtain 95.3%, 90.1%, 92.2% ratings respectively. The evaluated output result is presented in Table 3.5 when we used FastText word embedding for evaluating our results. Fig. 3.5 depict graphical rendition of our model's performance.

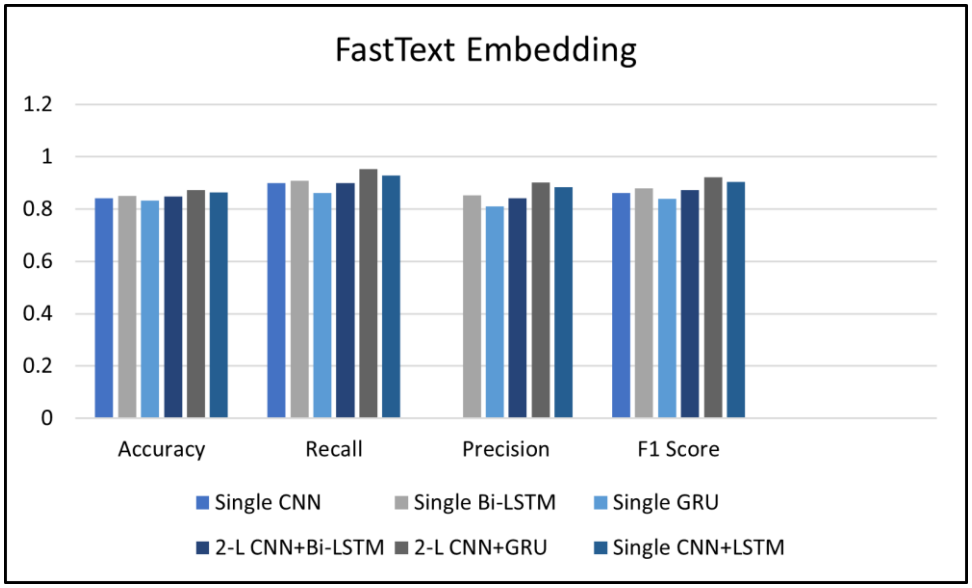

Fig. 3.5: Sentiment prediction with different layout of CNN, LSTM and GRU with FastText embedding

3.3 Comparative Analysis of Result and Discussion

For the traditional approach, it is obvious from the overall monitoring that Ensemble has been able to gain the highest accuracy for hotel review dataset. We also notice that, when we evaluate with the BoW technique, voting ensemble has been able to accomplish the highest accuracy. Furthermore, when we evaluate with TF-IDF the Bagging ensemble gives us the best result. So, for out review's dataset we can simply decide that the ensemble of different classifiers perform very well than simply using the classifier alone. Similarly, for the recall, precision and F1 score the ensemble model perform well in most of the cases. Again, the performance on the dataset assure that any underfitting or overfitting does not exist in our proposed approach. The recall, F1 score and precision does not change much and vary within a small range. The ensemble models perform well in most of the cases and getting better performance result.

In the Fig.3.6 we evaluate our traditional models based on feature extraction methods and accuracy.

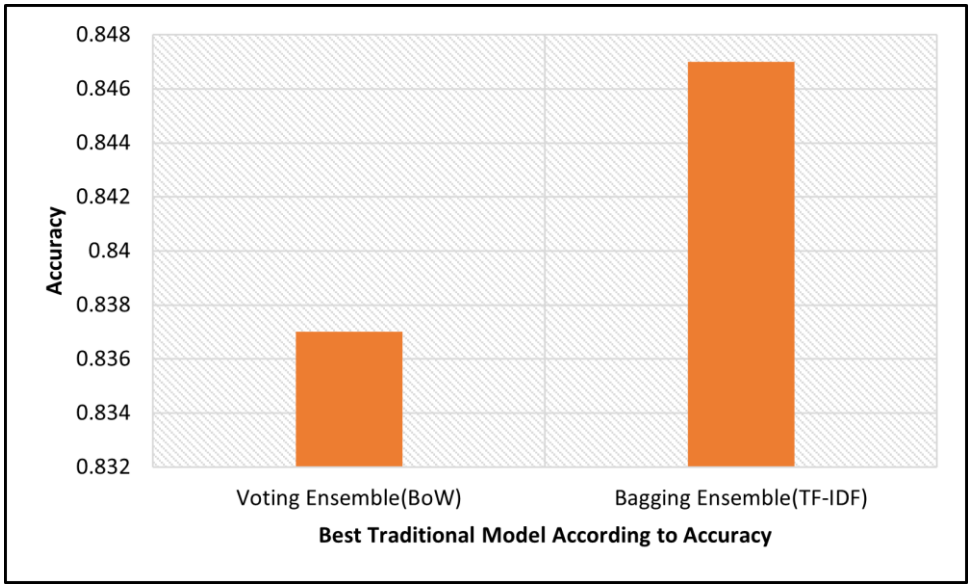

Fig. 3.6: Best Traditional Model According to Accuracy

For deep neural approach, it is obvious from the data-table and figure that the deep learning model perform the classification task more efficiently than the traditional classifier and even better than the ensemble models. The deep methods actually give the result that we have earned from the ensemble of multiple classifiers. Firstly, we use the embedding that is built-in in the keras model. There is no per-trained embedding in this case. The model learns from the scratch and this combination give efficient result than the traditional classifier. In every case, we use total 6 number of combinations namely single CNN, single Bi-LSTM, single GRU, 2-L CNN + Bi-LSTM, 2-L CNN + GRU and single CNN + LSTM. When we trained our data on the scratch, single CNN and LSTM networks has been able to accomplish the best result of accuracy. Then we examined with two different word embeddings namely GloVe and FastText respectively. When we trained our dataset on GloVe-300d, the single CNN and LSTM came up with the best accuracy result. Lastly, we applied the FastText embedding. When we train our data with FastText, the 2-L single CNN and GRU gives the most promising result. The ensemble models perform well in most of the cases and getting better performance result.

In the Fig.3.7 we define the best model according to accuracy based on the word embeddings.

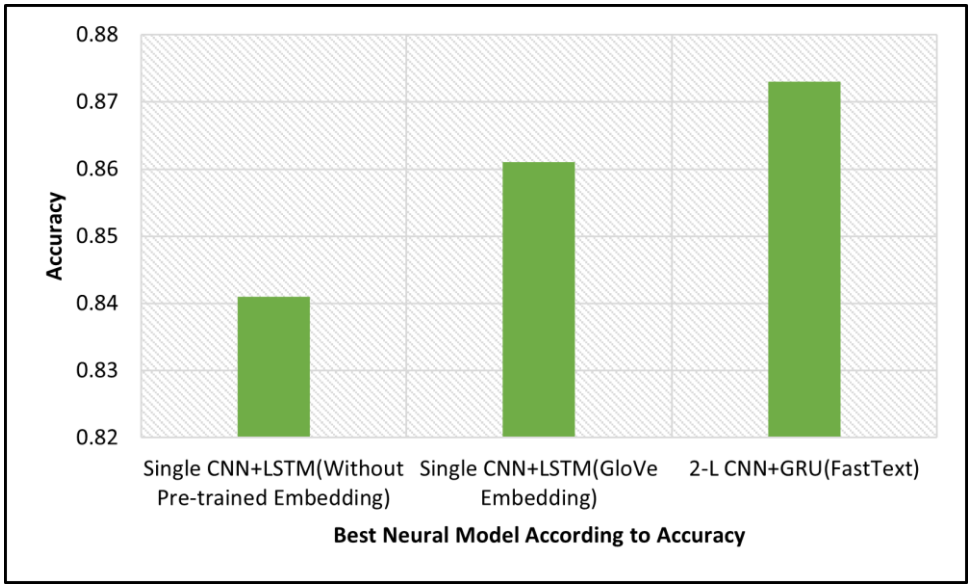

Fig. 3.7: Best Neural Model According to Accuracy

3.4 Conclusion

This chapter restrains our computational results. In the later phase of this chapter, we try to select the best model both for the traditional and deep learning based on the accuracy and we depict the generalized visualization of the best model.

Chapter 4

Conclusion & Future Work

4.1 Conclusion

Two feature extraction techniques (i.e., BoW, TF-IDF) were used on hotel review dataset in our research and a comparative analysis was shown to explore the potential of these techniques for text categorization through machine learning models (i.e., SVM, LR, Ensemble). In terms of accuracy, precision, recall and F1 score for binary opinion categorization, we suggest a strong approach to finding efficient function extraction and machine learning techniques. It is verified from the findings that strong performance for hotel review is obtained by both function extraction strategies and classifiers. The noteworthy findings reveal that the voting ensemble model offers the best performance for the BoW and bagging ensemble for the TF-IDF. Various layout of CNN, GRU and LSTM based methods of deep learning are checked for hotel review data sentiment analysis. This research suggested a profound learning model for an interpretation of binary opinion categorization, and the two-layer CNN-based model was skill fully paired with a LSTM-layers. FastText and GloVe word embedding models embrace this architecture as the input layer. The tests with a multi-domain corpus revealed that the model's efficiency was impressive in terms of precision, recall, F1 score and accuracy. This thesis thoroughly validated and concluded that the FastText embedding is a more important alternative in the learning of semantic and syntactic knowledge through word embedded techniques on the classification of hotel review data.

4.2 Future Work

We intend to try a broad twitter dataset in the future, and a lexicon-based approach will be applied to mark the dataset to conduct multi-class opinion classification. For future studies, the use of deep learning algorithms in the interests of user exploration and suggestion should be explored based on the interest. The efficiency of word embedding enhanced by combining the WordNet lexical database with the input layer.

References

[1] "An introduction to Bag of Words and how to code it in Python for NLP", *freeCodeCamp.org*, 2020. [Online]. Available: https: //www.google.com/ amp/s / www. freecodecamp.org /news/ an- introduction -to - bag- of - words- and- how- to- code - it- in – python – for – nlp - 282e87a9da04/ amp/ . [Accessed: 14- Dec- 2020].

[2] "TF IDF | TFIDF Python Example", *Medium*, 2020. [Online]. Available: https: //towardsdatascience.com/natural-language–processing–feature –engineering –using – tf – idf- e8b900e7e76. [Accessed: 14- Dec- 2020].

[3] "Understanding Convolutional Neural Networks for NLP", *WildML*, 2020. [Online]. Available: http:// www.wildml.com/ 2015/11/ understanding- convolutional - neural-networks-for-nlp/. [Accessed: 14- Dec- 2020].

[4] "Long Short-Term Memory: Akshay Sood | Computational Neuroscience | Algorithms", *Scribd*, 2020. [Online]. Available: https: //www.scribd.com/ document/ 461583872/Long-Short-Term-Memory-Networks-pdf. [Accessed: 14- Dec- 2020].

[5] "Understanding GRU Networks", *Medium*, 2020. [Online]. Available: https: //towardsdatascience.com/understanding-gru-networks-2ef37df6c9be. [Accessed: 15- Dec- 2020].

[6] D. Goularas and S. Kamis, "Evaluation of Deep Learning Techniques in Sentiment Analysis from Twitter Data", *2019 International Conference on Deep Learning and Machine Learning in Emerging Applications (Deep-ML)*, 2019. Available: 10.1109/deep-ml.2019.00011 [Accessed 14 December 2020].

[7] A. Ombabi, W. Ouarda and A. Alimi, "Deep learning CNN–LSTM framework for Arabic sentiment analysis using textual information shared in social networks", 2020. .

[8] P. Ray and A. Chakrabarti, "A Mixed approach of Deep Learning method and Rule-Based method to improve Aspect Level Sentiment Analysis", *Applied Computing and Informatics*, vol. --, no. --, 2020. Available: 10.1016/j.aci.2019.02.002 [Accessed 14 December 2020].

[9] S. Lai, L. Xu, K. Liu and J. Zhao, "Recurrent convolutional neural networks for text classification", *In Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence (AAAI'15).*, vol. , 22672273, 2015. [Accessed 14 December 2020].

[10] M. Cliche, "BB_twtr at SemEval-2017 Task 4: Twitter Sentiment Analysis with CNNs and LSTMs", *Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017)*, 2017. Available: 10.18653/v1/s17-2094 [Accessed 14 December 2020].

[11] X. Liu, F. Xu, Y. Sun, H. Zhang and Z. Chen, "Convolutional Recurrent Neural Networks for Observation-Centered Plant Identification", *Journal of Electrical and Computer Engineering*, vol. 2018, pp. 1-7, 2018. Available: 10.1155/2018/9373210.

[12] H. Xu, M. Dong, D. Zhu, A. Kotov, A. Carcone and S. Naar-King, "Text Classification with Topic-based Word Embedding and Convolutional Neural Networks", *Proceedings of the 7th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics*, 2016. Available: 10.1145/2975167.2975176 [Accessed 14 December 2020].

[13] A. Shewalkar, D. Nyavanandi and S. Ludwig, "Performance Evaluation of Deep Neural Networks Applied to Speech Recognition: RNN, LSTM and GRU", *Journal of Artificial Intelligence and Soft Computing Research*, vol. 9, no. 4, pp. 235-245, 2019. Available: 10.2478/jaiscr-2019-0006 [Accessed 14 December 2020].

[14] M. Cai and J. Liu, "Maxout neurons for deep convolutional and LSTM neural networks in speech recognition", *Speech Communication*, vol. 77, pp. 53-64, 2016. Available: 10.1016/j.specom.2015.12.003 [Accessed 14 December 2020].

[15] N. Dang, M. Moreno-García and F. De la Prieta, "Sentiment Analysis Based on Deep Learning: A Comparative Study", *Electronics*, vol. 9, no. 3, p. 483, 2020. Available: 10.3390/electronics9030483 [Accessed 14 December 2020].

[16] "Leveraging Deep Learning for Multilingual Sentiment Analysis - AYLIEN News API", *Aylien.com*, 2021. [Online]. Available: https://aylien.com/blog/leveraging-deep-learning-for-multilingual. [Accessed: 15- Dec- 2020].

[17] "515K Hotel Reviews Data in Europe", *Kaggle.com*, 2020. [Online]. Available: https:// www. kaggle.com/jiashenliu/515k-hotel-reviews-data-in-europe. [Accessed: 15- Dec- 2020].

[18] "Lemmatization Approaches with Examples in Python", *ML+*, 2020. [Online]. Available: https://www.machinelearningplus.com/nlp/lemmatization-examples-python/.[Accessed: 15- Dec- 2020].

[19] C. Python, "What is Word Cloud | Guide To Create A Word Cloud in Python", *Analytics Vidhya*, 2020. [Online]. Available: https://www.analyticsvidhya.com/blog/2020/10/word-cloud-or-tag-cloud-in-python/. [Accessed: 14- Dec- 2020].

[20] C. Liu, "A Top Machine Learning Algorithm Explained: Support Vector Machines (SVM) - KDnuggets", *KDnuggets*, 2020. [Online]. Available: https: //www.kdnuggets.com /2020/03/ machine-learning-algorithm-svm-explained.html. [Accessed: 14- Dec- 2020].

[21] "The Logistic Regression Algorithm", *Experfy Insights*, 2020. [Online]. Available: https: //www.experfy.com/blog/bigdata-cloud/the-logistic-regression-algorithm/. [Accessed: 14- Dec- 2020].

[22] J. Brownlee, "How to Develop Voting Ensembles With Python", *Machine Learning Mastery*, 2020. [Online]. Available: https://machinelearningmastery.com/voting-ensembles-with-python /#:~:text= A%20voting%20ensemble%20 (or%20a,model%20used %20in% 20t h e%20ensemble. [Accessed: 14- Dec- 2020].

[23] "Introduction to Bagging and Ensemble Methods | Paperspace Blog", *Paperspace Blog*, 2020. [Online]. Available: https://www.google.com/amp/s/blog.paperspace.com/bagging-ensemble-methods/amp/. [Accessed: 14- Dec- 2020].

[24] J. Brownlee, "What Are Word Embeddings for Text?", *Machine Learning Mastery*, 2020. [Online]. Available: https://machinelearningmastery.com/what-are-word-embeddings/. [Accessed: 14- Dec- 2020].

[25] "fastText - Facebook Research", *Facebook Research*, 2020. [Online]. Available: https: // research.fb.com/blog/2016/08/fasttext/. [Accessed: 14- Dec- 2020].